FUNCTIONAL DESIGN SPECIFICATION

FOR THE

- P16 POUCH MACHINE-

MAIN ORDER

30.427

Documentation Approval:

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Contents

1 Version history

Version	Date	Author	Description
0.1	2020-08-11	Svenja Klipstein	Draft

2 Scope

This document is valid for the following orders.

Order no.	Machine type	Machine no.
30427	PMT 360 KV	HH-2248.005

3 Abbreviations

Abbreviation	Explanation
AN	Customer
ASi	Actuator-Sensor-Interface (Feldbussystem)
BGV	Regulations of employer's association
CFR	Code of Federal Regulations
cGMP	current Good Manufacturing Practice
DIN	German Institute for standards (Deutsches Institut für Normung e.V.)
EMV	Electro-magnetic compatibility
EN	European Standard
FAT	Factory Acceptance Test
FDA	Food and Drug Administration
FIFO	First in First Out (memory organization principle)
FMEA	Failure Mode and Effect Analysis
GAMP	Good Automated Manufacturing Practice
GSG	Appliance safety regulation
HF	High Frequency
HMI	Human Machine Interface
IEC	International Electrotechnical Commission
IPC	In-Process Control or Industry-PC
IQ	Installation Qualification
NFPA	National Fire Protection Agency

Abbreviation	Explanation
OPC	Open Process Communication Commission
OQ	Operational Qualification
SCADA	Supervisory Control and Data Acquisition (Linienleitsystem)
SOP	Standard Operation Procedure (operating instruction)
UL	Underwriters' Laboratory (American safety and standards organization)
VDE	Association of electrical technology, electronics and information technology
XML	Extended Markup Language (Communication format standard)

4 Units

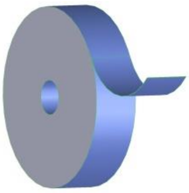
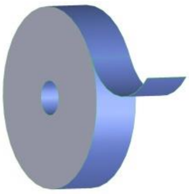
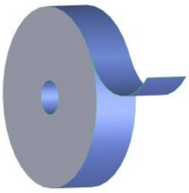
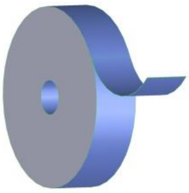
Abbreviation	Explanation
N	Newton
mg	Milligram
g	Gram
kg	Kilogram
ms	Millisecond
s	Second
min	Minutes
h	Hour
µm	Micrometer
mm	Millimeter
m	Meter
°C	Degrees Celsius
V	Volt
Hz	Herz
bar	Bar
dB	Decibel

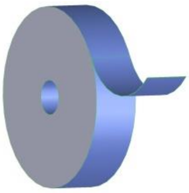
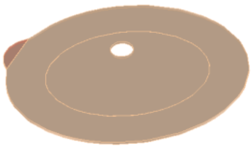
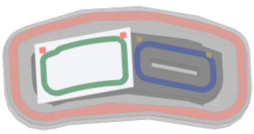
5 Introduction

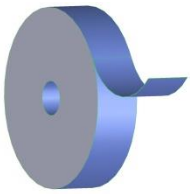
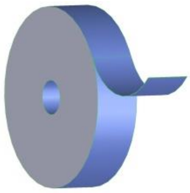
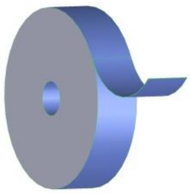
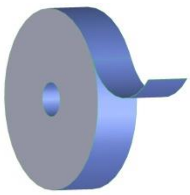
The machine is designed to:

- Unwinding NW BS
- Moon punch NW BS
- Unwinding Saranex BS
- Corona treatment BS
- Thermo Printer Saranex BS
- Print verification
- Tack sealing BS
- Backing sealing
- Moon punch Saranex BS
- Feeding PAD
- Sealing PAD to Saranex BS
- PAD verification
- Unwinding Saranex NBS
- Corona treatment NBS
- Feeding flap
- Sealing flap
- Punch filter hole
- Feeding filter
- Sealing filter
- Filter verification
- Unwinding VO film and NW NBS
- VO smash sealing
- VO cut
- VO water trap sealing
- Unwinding NW NBS
- Tack sealing BS/NBS films
- Peripheral seal
- Feeding Velour
- Feeding endstrip NBS
- Feeding endstrip BS
- Label verification
- Heating and pressing label
- Peripheral punch
- Discharge bad parts
- Pick and place good bags onto discharge belt
- Reject waste grid

6 Components / Materials to be processed

Film BS 	Manufacturer: xxx Material: CFF2, beige and clear Sample available? Yes ☒ / No ☐ core diameter: 76,2mm (3") diameter: max. 700mm actual: xxxmm weight: max. 100kg acutal: xxxkg width: 320mm
Film NBS 	Manufacturer: xxx Material: CFF2, beige and clear Sample available? Yes ☒ / No ☐ core diameter: 76,2mm (3") diameter: max. 700mm actual: xxxmm weight: max. 100kg acutal: xxxkg width: xxxmm
Film VO 	Manufacturer: xxx Material: MO film, clear Sample available? Yes ☒ / No ☐ core diameter: 76,2mm (3") diameter: max. 700mm actual: xxxmm weight: max. 100kg acutal: xxxkg width: 60mm
NW BS 	Manufacturer: xxx Material: RKW Dansac, beige Sample available? Yes ☒ / No ☐ core diameter: 76,2mm (3") diameter: max. 1000mm actual: xxxmm weight: max. 100kg acutal: xxxkg width: xxmm

NW NBS 	Manufacturer: xxx Material: RKW Dansac, beige Sample available? Yes ☒ / No ☐ core diameter: 76,2mm (3”) diameter: max. 1000mm actual: xxxmm weight: max. 100kg acutal: xxxkg width: xxmm
Barrier 	Material: film (pouch side), hydrocolloid, film Sample available? Yes ☒ / No ☐
Gasket 	Material: material, clear Sample available? Yes ☒ / No ☒
Soft convex 	Material: material, clear Sample available? Yes ☒ / No ☐
Convex 	Material: material, clear Sample available? Yes ☒ / No ☐
Filter AF 300 	Material: material, clear Sample available? Yes ☒ / No ☐ core diameter: 76,2mm (3”) diameter: max. 800mm actual: 800mm weight: max. 100kg actual: xxxkg width: 81,7mm

Flap 	Manufacturer: xxx Material: material, beige Sample available? Yes ☒ / No ☐ core diameter: 76,2mm (3") diameter: max. 800mm actual: xxxmm weight: max. 100kg acutal: xxxkg width: xxmm
Velour 	Manufacturer: xxx Material liner: xxx, clear Material member: xxx Sample available? Yes ☒ / No ☐ core diameter: 76,2mm (3") diameter: max. 450mm actual: xxmm 8.900 members per roll @ pitch of 11mm weight: max. 100kg actual: xxxkg
Endstrip BS 	Manufacturer: xxx Material liner: paper, beige Material member: xxx Sample available? Yes ☒ / No ☐ core diameter: 76,2mm (3") diameter: max. 400mm actual: xxmm 9.000 strips per roll @ pitch of 17mm weight: max. 100kg actual: xxxkg
Endstrip NBS 	Manufacturer: xxx Material liner: paper, beige Material member: xxx Sample available? Yes ☒ / No ☐ core diameter: 152,4mm (6") (rounded side leading edge first) diameter: max. 400mm actual: xxmm 9.000 strips per roll @ pitch of 17mm weight: max. 100kg actual: xxxkg

7 Functional Description

7.1 Operational sequence

Unwinding Saranex BS
▼
Splice detection Saranex BS
▼
Web guiding Saranex BS
▼
Corona treatment BS
▼
Pull roller Saranex BS
▼

Unwinding NW BS		Thermo printer Saranex BS
▼		▼
Web guiding NW BS		Print verification
▼		▼
Splice detection NW BS		Web tension verification Saranex BS
▼		▼
Pull roller NW BS		Web guiding Saranex BS
▼		▼
Moon punch NW BS	►	Tack sealing BS

▼
Pull roller BS
▼
Web tension verification BS
▼
Chain index system
▼
A

A		
▼		
Backing sealing		
▼		
Moon punch Saranex BS		
▼		
PAD sealing station		
▼		
PAD verification	◄	PAD transfer
▼		▲
B		PAD detection
		▲
		Double stack detection
		▲
		PAD conveyor
		▲
		Gasket singulation
		▲
Barrier singulation	►	Gasket magazine
▲		
Barrier detection		
▲		
Barrier infeed conveyor		

		Unwinding Saranex NBS		
		▼		
		Web guiding Saranex NBS		
		▼		
		Splice detection Saranex NBS		
		▼		
		Pull roller Saranex NBS		
		▼		
		Corona treatment NBS		
		▼		
		Web tension verification Saranex NBS		
		▼		
		Pull roller Saranex NBS 2		
		▼		
		Web tension verification Saranex NBS 2		
		▼		
Unwinding flap	►	Feeding and sealing flap	►	Waste grid removal flap
		▼		
		Punch filter hole		
		▼		
Punch and transfer filter	►	Feeding and sealing filter	►	Waste grid removal filter
▲		▼		
Pull roller filter		Filter verification		
▲		▼		
Vision system reject marking		**C**		
▲				
Splice detection filter				
▲				
Unwinding filter				

Unwinding VO film
▼
Web guiding VO film
▼
Pull roller VO film
▼
Splice detection VO film
▼
Web tension verification VO film ►

Unwinding NW NBS
▼
Web guiding NW NBS
▼
Pull roller NW NBS
▼
◄ Splice detection NW NBS

VO smash sealing
▼
VO cut
▼
Pull roller VO
▼
Web tension verification VO
▼
C ► VO water trap sealing
▼
Pull roller NBS
▼
Web guiding NBS film
▼
Web tension verification NBS
▼
D

		B	◄	**D**
		▼		
Unwinding Velour		Tack sealing BS/NBS films		
▼		▼		
Feeding Velour		Peripheral seal		Unwinding endstrip NBS
▼		▼		▼
Detection Velour	►	Transfer Velour		Feeding endstrip NBS
▼		▼		▼
Waste grid removal Velour		Transfer endstrip NBS	◄	Detection endstrip NBS
		▼		▼
Waste grid removal endstrip BS		▼		Waste grid removal endstrip NBS
▲		▼		
Detection endstrip BS	►	Transfer endstrip BS		
▲		▼		
Feeding endstrip BS		Label verification		
▲		▼		
Unwinding endstrip BS		Heating and pressing label		
		▼		
		Peripheral punch		
		▼		
Pull rolls waste grid	◄	Discharge belt bags		
		▼		
Reject bad bags	◄	Pick and place bags		
		▼		
		Discharge belt good parts		

7.2 Technical data

Description	Data
Output:	Average output: 22 cpm (44 ppm) Design speed: 24 cpm (48 ppm)
Cycle time:	2,7 s
Noise level in accordance with EN ISO 37441-46 (DIN 45635 T. 3)	LPA < 78 dB(A)
Weight	See identification plate
Dimensions (length x width x height):	See machine layout
Machine color:	HH-standard-7955 silver metallic, satin (S431) VA -look similar to RAL9007
Lubricants not covered areas:	FDA approved
Steel components:	Electro-plated with nickel
Aluminum components:	Anodized / hard-anodized
Rubber components	e.g. Silicone
Plastic components	e.g. Delrin
Special materials	e.g. G60, Pyropel MD 12
Special coatings:	e.g. Teflon coating
Product contact parts <u>metal</u>	ASI 304 aluminum hard-anodized Aluminum Teflon coated
Product contact parts <u>rubber</u>	FDA approved, Latex free
Product contact parts <u>plastic</u>	FDA approved
Conformation of product contact parts	Parts list 2.1 certificate

7.3 Functional description of the individual stations

7.3.1 General Machine Design

The base machine consists of:

Denomination	Execution		Comment
Frame	Working height: see layout		n.a.
	Clearance between machine frame and floor: Minimum 100 mm		Design: 125± 25mm CAD-110mm
	Machine wall, aluminum anodized.		n.a.
	Welded steel frame, lacquer-coated.		n.a.
	Steel frame is divided in 5 sections which are bolted and pined together		n.a.
	The product section is separated from the drive and installation sections.		n.a.
	Stations mounted via linear sliding rails to machine wall		n.a.
	Lubrication points connected to lubrication nipples on central plates, easy accessible, with color marking for necessary relubrication time.		n.a.
	Control panel height max. 1500 mm.		n.a.
	Description/Data		
	Dimensions Base machine in mm (L x W x H):	According layout M12045	n.a.
Transport system	**Execution**		
	Chain index system		n.a.
	Description/Data		
	Method of operation	Intermediate motion	n.a.
	Pitch station	163mm	n.a.
	Machined parts per station	2	n.a.
	Drive	Servo motor	n.a.
Utility Supply	**Execution**		
	n.a.		n.a.
	Description/Data		
	Power supply: 400 V AC (± 10%), 3-phase, N/PE, 50 Hz (± 1%) Control voltage 24 V DC Compressed air, min 87 PSI (6 bar), oil-free with filter		Power supply is connected to switch cabinet.
Signaling and	**Execution**		

Denomination	Execution	Comment
Controls	Stack light ➢ Intermittent red light: Illuminated if equipment is stopped due to any alarm. ➢ Intermittent yellow light: Illuminated if product level sensor is activated. ➢ Continuous green light: Illuminated if the equipment is running normally. ➢ Intermittent green light: Illuminated to signal that machine is about to start operation. An audible horn shall be activated when the machine is about to start.	n.a.
	Control panel ➢ Switches: Start, Stop, Reset.	n.a.

7.3.1.1 Pneumatics

Designation	Execution		Comments
Pneumatic Components	Service Unit Cylinders Valves / Valve Stations	(Festo) (Festo) (Festo)	n.a.
Vacuum Components	Side channel compressor Valves / Valve Stations	(Becker) (Schmalz / Festo)	n.a.
Colours of Utility Hoses	Blue Silver Transparent Black and red insolated hose Red	Compressed Air (6 bar) Control Air (6 bar) Vacuum Cooling Water Continuous Air (6 bar)	n.a.

7.3.1.2 Guards

Locked guard doors ensure that there are no interventions by the operator while the machine is running.

It is not possible to start the machine if one guard door is open.

Designation	Execution	Comments
Guards	Guard door execution as wing doors and folding doors. Height of guards - see layout. Distance between guard doors and the floor – see layout Guard doors are interlocked during automatic operation For a better cleaning there are no holes in the guard doors. Open guard doors are identified on the HMI.	n.a.
	Description/Data	
Function parameters	n.a.	
Programming instructions	n.a.	

7.3.1.3 Interface to upstream machine

Designation	Execution		Comments
	n.a.		n.a.
	n.a.		n.a.
	n.a.		n.a.
	Description/Data		
	n.a.	n.a.	n.a.
	n.a.	n.a.	n.a.
	n.a.	n.a.	n.a.
Programming instructions		n.a.	

7.3.1.4 Interface to downstream machine

Designation	Execution		Comments
	n.a.		n.a.
	n.a.		n.a.
	n.a.		n.a.
	Description/Data		
	n.a.	n.a.	n.a.
	n.a.	n.a.	n.a.
	n.a.	n.a.	n.a.
Programming instructions		n.a.	

7.3.1.5 General Electrical Design

Designation	Description/Data	Comments
Electrical Design **General**	All electrical configurations fulfill the requirements in the machinery directive.	n.a.
Electrical Design	The electrical equipment / installation are designed according to EN 60204-1 safety requirements.	n.a.
Electrical Design	Control voltages: 24 V DC. Field Bus for Communication: Profibus, ASi, ASi-Safety	n.a.
Electrical Design	OiT/HMI realized with 15” Touch-Screen, I-PC based.	n.a.
Electrical Design	Electrical installation acc. EMC - guideline.	n.a.

7.3.2 Stations

7.3.2.1 Station 1010 und 1014: Moon punch BS

A hole is punched through the BS material. For punching a steel rule die with an anvil plate is used. The punching waste is sucked into a waste container.

The moon punch station is driven by a servo motor via an eccentric shaft.

Designation	Execution		Comments
punch	Eccentric shaft is driven by a servo motor via gearing and coupling		n.a.
	Adjusting unit for automatic Y-adjustment via servo motor		n.a.
	X-adjustment manually via linear guides		n.a.
	Description/Data		
	sensors	See technic. doc.	n.a.
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
extraction	Extraction is tuned on / off by valve		n.a.
	Vacuum removal of punching waste		n.a.
	Description/Data		
	valve	See technic. doc.	n.a.
Format parts	Punching tool is a steel rule die		n.a.
	Steel rule die and anvil plate are held by spring load (manually locked by closing disc)		n.a.
	Anvil plate is hardened HRC 62+1, Ra0,1 polished, AlTi coated		n.a.
	Plunger and suction funnel depend on format		n.a.
	Plunger and suction funnel are fixed by an index bolt		n.a.
	Format parts are designed for Rapid Change Over		n.a.
	Opening and closing of tool clamping by cylinders via ASI module with opened guard doors		n.a.
	Description/Data		
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.2 Station 1012 Chain index system

The BS film and BS nonwoven material is feed into the machine by pull rolls.

BS film is gripped and indexed by a gripper chain.

Designation	Execution		Comments
Chain index system	Grippers are opened at beginning and end of chain		n.a.
	Chain is driven by a servo motor via gearing and coupling		n.a.
	Chain tension is adjusted mechanically		n.a.
	Chain guide is made of s-green		n.a.
	Chain guide is mounted to machine wall by aluminum profiles		n.a.
	Distance between chains and chain orientation is adjustable (one time adjustment) to adjust the material tension		n.a.
	Always change both chains (never just one side)		n.a.
	Description/Data		
	chain pitch	5/8”	n.a.
	advance time	approx. 0,7 s	n.a.
	advance	See technic. doc.	n.a.
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.3 Station 1013: Backing sealing

The cut non-woven is sealed to the BS film.

Designation	Execution		Comments
Sealing	Down holder from top driven by cylinder		n.a.
	Sealing unit from bottom driven by cylinder		n.a.
	Heating plate with pluggable heating elements and PT 100		n.a.
	Sealing pressure adjustable via proportional valve		n.a.
	Heating plate isolated from the outside (G60)		n.a.
	Adjusting unit for automatic Y-adjustment via servo motor		n.a.
	X-adjustment manually via linear guides		n.a.
	Description/Data		
	sealing time	Max. 1,0s	n.a.
	sealing temperature		n.a.
	sealing pressure		n.a.
	heating elements	See technic. doc.	n.a.
	PT 100	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
	servo motor and gear	See technic. doc.	n.a.
Format parts	Down holder and back up plate held by spring load		n.a.
	Format parts are designed for Rapid Change Over		n.a.
	Opening and closing of down holder clamping by cylinders via ASI module with opened guard doors		n.a.
	Description/Data		
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.4 Station 1016: PAD sealing

PADs are fed into sealing position and sealed to the film BS.

Designation	Execution		Comments
Sealing	Sealing tool driven independently by servo motor via electrical cylinder		n.a.
	Fixture with back up tool is lifted upwards by servo motor via eccentric shaft		n.a.
	Sealing tool with pluggable heating elements and PT 100		n.a.
	Sealing force controlled by load cells		n.a.
	Sealing tool isolated from the outside (G60)		n.a.
	Adjusting unit for automatic Y-adjustment via servo motor		n.a.
	Sealing tools are adjustable in Y-direction to align sealing tool to fixture		n.a.
	X-adjustment manually via linear guides		n.a.
	Back up tool with self leveling feature		n.a.
	Description/Data		
	sealing time	max. 1,0 s	n.a.
	sealing temperature		n.a.
	sealing force		n.a.
	heating elements	See technic. doc.	n.a.
	PT 100	See technic. doc.	n.a.
	Servo motors	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
Rotatory table	Rotatory table with two fixtures		n.a.
	In one fixture the PADs are placed and on the other fixture the PADs are sealed		n.a.
	Rotatory table is driven by servo motor via gearing		n.a.
	Rotation 180°		n.a.
	Description/Data		
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
Teflon belt	Manual movement of Teflon belt into or out of sealing area possible; fixed in both positions by ratchet		n.a.
	Not actively driven		n.a.
	Tension on Teflon belt can be adjusted mechanically		n.a.

	Tension is held and released by pneumatic cylinder		n.a.
	Description/Data		
	cylinder	See technic. doc.	n.a.
	valve	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
Format parts	PADs are placed into fixture and are sealed to film BS		n.a.
	Convex parts and base plates are held on fixture by vacuum		n.a.
	Gaskets are centered in fixture and ejected by cylinder		n.a.
	Sealing tool and down holder are fixed by captive screws		n.a.
	Fixture mounted on rotatory table by quick release fastener		n.a.
	Format parts are designed for Rapid Change Over		n.a.
	Description/Data		
	cylinder	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	limit switch	See technic. doc.	n.a.
Programming instructions		Calibration cycle necessary Multicam necessary	

7.3.2.5 Station 1019: PAD verification

The PAD verification is done by a camera system. The system indicates the center and rotation of the barrier or gasket. Additional features see Vision FDS.

Due to the need of different lighting for barriers and gaskets the lights need to move.

Designation	Execution		Comments
PAD verification	Lights are moved by servo via toothed belt axis due to the need of different lighting for barriers and gaskets		n.a.
	Particles that may be on the glass of the camera are blown off by air		n.a.
	Camera is mounted underneath the chain to have a good view on the PAD		n.a.
	X-adjustment manually via linear guides		n.a.
	Description/Data		
	servo motor	See technic. doc.	n.a.
	vision details	See vision FDS	n.a.
	sensor	See technic. doc.	n.a.
	valve	See technic. doc.	
Programming instructions		n.a.	

7.3.2.6 Station 1032: Thermo printer

Film BS is printed by the two thermo printer.

Designation	Execution		Comments
Printer	printer is printing the film BS		n.a.
	Printers can be moved manually in Y direction and fixed via ratchet		n.a.
	Positioning of printer is detected by sensor		n.a.
	Idle rollers are adjustable		n.a.
	Description/Data		
	Thermo printer	Type: Markem Smart Date X40	n.a.
	sensors	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.7 Station 1033: Print verification

The print verification is done by a camera system.

Additional features see Vision FDS.

Designation	Execution		Comments
Filter verification	Light is mounted on top of film and camera underneath		n.a.
	X-adjustment manually via linear guides		n.a.
	Description/Data		
	Vision details	See vision FDS	n.a.
Programming instructions		n.a.	

7.3.2.8 Station 1037: Feeding and sealing flap

The web is feed into the punch where the flap is punched out of the web. The flap is picked about of the punch and transferred onto the backup sealing pad. The flap is moved into sealing position by the backup sealing pad. Flap is sealed to the NBS film.

Designation	Execution		Comments
Basic station	X-adjustment manually via linear guides		n.a.
	Description/Data		
	Servo motor / gear	See technic. doc.	n.a.
Punching	Driven by cylinders		n.a.
	X-adjustment manually via linear guides		n.a.
	Complete punching unit is held by spring load (manually locked by closing disc)		n.a.
	Punching tool is a male – female tool		n.a.
	Punching tool is designed for Rapid Change Over		n.a.
	Opening and closing of tool clamping by cylinder via ASI module with opened guard doors		n.a.
	Description/Data		
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
Transfer	The punched flap is picked up and placed on the portal by rotatory cylinder		n.a.
	The rotary unit is lifted by a cylinder		n.a.
	The portal is driven by a servo motor via gearing coupling and toothed belt axis		n.a.
	Description/Data		
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
Sealing	Sealing from below the film, driven by cylinders		n.a.
	Fixture of portal is moved downwards by a cylinder		n.a.
	Heating plate with pluggable heating elements and PT 100		n.a.
	Sealing pressure adjustable via proportional valve		n.a.
	Heating plate isolated from the outside (G60)		n.a.

	Sealing tools are adjustable in Y-direction via servo motor		n.a.
	Sealing units are designed for Rapid Change Over		n.a.
	Description/Data		
	sealing time	max. 1,0s	n.a.
	sealing temperature		n.a.
	sealing pressure		n.a.
	heating elements	See technic. doc.	n.a.
	PT 100	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
	Servo motor / gear	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.9 Station 1038 Punch filter hole

Film NBS is punched to allow the gas to escape the bag through the filter.

Designation	Execution		Comments
Punch	Punching tool as male female die		n.a.
	Punch is driven by cylinder		n.a.
	Ejector and suction funnel to remove punching waste		n.a.
	Adjusting unit for automatic Y-adjustment via servo motor		n.a.
	X-adjustment manually via linear guides		n.a.
	Description/Data		
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
	ejector	See technic. doc.	n.a.
	servo motor / gear	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.10 Station 1040: Feeding and sealing filter

Filters are rotated into sealing position and sealed to the film NBS.

Designation	Execution		Comments
Sealing	Sealing tools are driven independently via cylinders		n.a.
	Fixture is lifted upwards in sealing position via cylinder		n.a.
	Sealing tool with pluggable heating elements and PT 100		n.a.
	Sealing tools isolated from outside (G60)		n.a.
	Adjusting unit for automatic Y-adjustment via servo motor		n.a.
	Sealing tools are adjustable in Y-direction to align sealing tool to fixture		n.a.
	X-adjustment manually via linear guides		n.a.
	Description/Data		
	Sealing time	max. 1,0s	n.a.
	Sealing temperature		
	Heating elements	See technic. doc	n.a.
	PT 100	See technic. doc	n.a.
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
	servo motor / gera	See technic. doc.	n.a.
Rotatory table	Rotatory table with two fixtures		n.a.
	In one fixture the filters are placed and on the other fixture the filters are sealed		n.a.
	Rotatory table is driven by servo motor via gearing		n.a.
	Rotation 180°		n.a.
	Description/Data		
	Servo motor	See technic. doc.	n.a.
	Gearing	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
Format parts	Filters are held on fixture by vacuum		n.a.
	Sealing tool is looked with ratchet		n.a.
	Fixture mounted on rotary table by spring load		n.a.
	Opening and closing of tool clamping by cylinders via ASI module with opened guard doors		n.a.
	Format parts are designed for Rapid Change Over		n.a.
	Description/Data		
	cylinder	See technic. doc.	n.a.

	valves	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.11 Station 1042: Filter verification

The filter verification is done by a camera system. The system indicates the position and the rotation of the filter.

Designation	Execution		Comments
Filter verification	Light is mounted on top of film and camera underneath		n.a.
	X-adjustment manually via linear guides		n.a.
	Description/Data		
	Vision details	See vision FDS	n.a.
Programming instructions		n.a.	

7.3.2.12 Station 1054: VO smash sealing

The non-woven of the NBS is smash sealed to the VO film.

Designation	Execution		Comments
Sealing	Down holder from bottom driven by cylinder		n.a.
	Sealing unit from top driven by cylinder		n.a.
	Heating plate with pluggable heating elements and PT 100		n.a.
	Sealing pressure adjustable via proportional valve		n.a.
	Heating plate isolated from the outside (G60)		n.a.
	Adjusting unit for automatic Y-adjustment via servo motor		n.a.
	X-adjustment manually via linear guides		n.a.
	Description/Data		
	sealing time	Max. 1,0s	n.a.
	sealing temperature		n.a.
	sealing pressure		n.a.
	heating elements	See technic. doc.	n.a.
	PT 100	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
	Servo motor / gear	See technic. doc.	n.a.
Format parts	Down holder and counter plate held by spring load		n.a.
	Format parts are designed for Rapid Change Over		n.a.
	Opening and closing of down holder clamping by cylinders via ASI module with opened guard doors		n.a.
	Description/Data		
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.13 Station 1056: VO cut

The viewing cut is made into the VO film and the non-woven. For punching a steel rule die with an anvil plate is used.

The VO punch station is driven by a servo motor via an eccentric shaft.

Designation	Execution		Comments
Punch	Eccentric shaft is driven by a servo motor via gearing and coupling		n.a.
	Adjusting unit for automatic Y-adjustment via servo motor		n.a.
	X-adjustment manually via linear guides		n.a.
	Description/Data		
	sensors	See technic. doc.	n.a.
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
Format parts	Punching tool is a steel rule die		n.a.
	Steel rule die and anvil plate are held by spring load (manually locked by closing disc)		n.a.
	Anvil plate is hardened HRC 62+1, Ra0,1 polished, AlTi coated		n.a.
	Format parts are designed for Rapid Change Over		n.a.
	Opening and closing of tool clamping by cylinders via ASI module with opened guard doors		n.a.
	Description/Data		
	steel rule die	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.14 Station 1061: VO water trap sealing

The VO film is sealed to the NBS film.

Designation	Execution		Comments
Sealing	Down holder from bottom driven by cylinder		n.a.
	Sealing unit from top driven by cylinder		n.a.
	Heating plate with pluggable heating elements and PT 100		n.a.
	Sealing pressure adjustable via proportional valve		n.a.
	Heating plate isolated from the outside (G60)		n.a.
	Adjusting unit for automatic Y-adjustment via servo motor		n.a.
	X-adjustment manually via linear guides		n.a.
	Description/Data		
	sealing time	Max. 1,0s	n.a.
	sealing temperature		n.a.
	sealing pressure		n.a.
	heating elements	See technic. doc.	n.a.
	PT 100	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
	Servo motor / gear	See technic. doc.	n.a.
Format parts	Down holder and counter plate held by spring load		n.a.
	Format parts are designed for Rapid Change Over		n.a.
	Opening and closing of down holder clamping by cylinders via ASI module with opened guard doors		n.a.
	Description/Data		
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.15 Station 1064: Peripheral seal

Final sealing of all material layers in the peripheral sealing station.

Designation	Execution		Comments
Sealing	Sealing tool top driven by servo motor via belt		n.a.
	Sealing tool bottom driven by servo motor via coupling		n.a.
	Sealing tool with pluggable heating elements and PT 100		n.a.
	Sealing pressure adjustable via servo drive		n.a.
	Sealing pressure controlled by load cell		n.a.
	Adjusting unit for automatic Y-adjustment via servo motor		n.a.
	X-adjustment manually via linear guides		n.a.
	Description/Data		
	sealing time	max. 1,0s	n.a.
	sealing temperature		n.a.
	servo motor	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
Format parts	Sealing tool with pluggable heating elements and PT 100		n.a.
	Down holder, top sealing tool and bottom sealing tool are held by spring load		n.a.
	Heating plate isolated from the outside (G60)		n.a.
	Sealing tool isolated (Pyropel MD12)		n.a.
	Opening and closing of tool clamping by cylinders via ASI module with opened guard doors		n.a.
	Format parts are designed for Rapid Change Over		n.a.
	Description/Data		
	heating elements	See technic. doc.	n.a.
	PT 100	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
Programming instructions		Calibration cycle necessary Multicam necessary	

7.3.2.16 Station 1066 and 1068: Corona treatment

The corona treatment improves the adhesion of the labels to the film.

Treated surfaces are on the BS and NBS film where the labels will be applied.

Designation	**Execution**		**Comments**
Corona	2up Corona unit Supplier of the corona system: MK automation ApS Supplier of corona components: Vetafone / Markro		n.a.
	A pulsed vacuum system in the cooling (back-up) plate that ensures the film to be treated is flat against the plate. An extraction unit allows Ozone generated by the treatment to be exhausted to atmosphere. The exhaust will likely be through the factory roof. The extraction flow must be adjustable to prevent the extraction from pulling the film against the electrode.		n.a.
	A pulsed treatment that allows for an initial pulse of corona, followed by a short pause, followed again by another pulse of corona. This prevents overheating of the film. HMI control of the corona system for: Duration of 0.35s on / 0.10s off / 0.35s on +/- 0.5 seconds Power of 0 to 400 W +/- xx		n.a.
	Adjusting unit for automatic Y-adjustment via servo motor		n.a.
	X-adjustment manually via linear guides		n.a.
	Description/Data		
	corona unit	See technic. doc.	n.a.
	cylinder	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
	valve	See technic. doc.	n.a.
	Servo motor / gear	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.17 Station 1069: Label verification

The label verification is done by a camera system. The system indicates the position and the rotation of the filter

Designation	Execution		Comments
Filter verification	Light is mounted on top of film and camera underneath		n.a.
	X-adjustment manually via linear guides		n.a.
	Description/Data		
	Vision details	See vision FDS	n.a.
Programming instructions		n.a.	

7.3.2.18 Station 1072: Heating and pressing label

The applied labels are pressed to the film for a better adhesion. The process is supported by heat.

Designation	Execution		Comments
Basic station	Station covers 3 indexes to achieve sufficient time for the process (activation of glue)		n.a.
	Adjusting unit for automatic Y-adjustment via servo motor		n.a.
	X-adjustment manually via linear guides		n.a.
	Description/Data		
	limit switches	See technic. doc.	n.a.
	Sevo motor / gear	See technic. doc.	n.a.
Heating / Pressing	Pressing with cylinders from top and bottom		n.a.
	BS and NBS end strips handeled together as one pressing unit		n.a.
	Velour strip haneled as an independent unit		n.a.
	Heat applied from top and bottom side		n.a.
	Heating tool with pluggable heating elements and PT 100		n.a.
	Description/Data		
	cylinders	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
	heating time per index	Up to. 1,5s per index	n.a.
	heating time in total	3-4,5 sec	n.a.
	temperature	60°C	n.a.
Programming instructions		n.a.	

7.3.2.19 Station 1076: Peripheral punch

Final punch of the bags and transfer onto the discharge belt.

Designation	Execution		Comments
Punch	Eccentric shaft is driven by a servo motor via gearing and coupling		n.a.
	Adjusting unit for automatic Y-adjustment via servo motor		n.a.
	X-adjustment manually with position control		n.a.
	Description/Data		
	sensors	See technic. doc.	n.a.
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
Ejector	Ejector with vacuum pushes the punched bag through the station and places the bag on the conveyor		n.a.
	Vacuum pump for vacuum supply		n.a.
	Description/Data		
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
	vacuum pump	See technic. doc.	n.a.
Format parts	Punching tool is a steel rule die		n.a.
	changing suction cup from open to closed pouch automatically via cylinder		n.a.
	Steel rule die and anvil plate are held by spring load (manually locked by closing disc)		n.a.
	Anvil plate is hardened HRC 62+1, Ra0,1 polished, AlTi coated		n.a.
	Format parts are designed for Rapid Change Over		n.a.
	Frame for lifter to change steel rule die		n.a.
	Opening and closing of tool clamping by cylinders via ASI module with opened guard doors		n.a.
	Description/Data		
	steel rule die	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.20 Station 1078: Discharge belt bags

The discharge belt transfers the bags from the peripheral punch station to the pick and place station. Bags which are not picked up by the pick and place station are bad parts and remain on the belt.

The bad bags are transferred into a waste container. The waste container can emptied during the machine is running.

Designation	Execution		Comments
Conveyor	Discharge belt for transfer bags to next station		n.a.
	Driven by servo motor via chain		n.a.
	Bad parts remain on discharge belt and transferred into a waste container		n.a.
	Description/Data		
	servo motor	See technic. doc.	n.a.
	conveyor	See technic. doc.	n.a.
Waste container	Container is held in place via magnetic switches		n.a.
	Extract container manually with support of pneumatic cylinder		n.a.
	Description/Data		
	cylinder	See technic. doc.	n.a.
	valve	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.21 Station 1080: P&P bags

The good bags are picked up and placed on the discharge belt good parts by the pick and place unit.

Designation	Execution		Comments
Pick and place	Vacuum gripper mounted on linear guiding		n.a.
	Spreading of the grippers by cylinder		n.a.
	Linear movement driven by a servo motor via gearing and coupling		n.a.
	Vacuum pump for vacuum supply		n.a.
	Valve to turn on or off the vacuum for single suction cup which is only needed for open pouches		n.a.
	Description/Data		
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
	vacuum pump	See technic. doc.	n.a.
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.22 Station 1082: Discharge belt good bags

Two discharge belts transferred the good bags out of the machine.

Bags can be imbricated in different styles.

Designation	Execution		Comments
Conveyor	Discharge belt for transferring bags out of machine		n.a.
	Driven by servo motor via chain		n.a.
	Bags can be imbricated in different styles		n.a.
	Advance is adjustable		n.a.
	Description/Data		
	servo motor	See technic. doc.	n.a.
	conveyor	See technic. doc.	n.a.
	height of conveyor	825 mm	n.a.
Programming instructions		n.a.	

7.3.2.23 Station 1084: Waste grid removal

The waste grid is pulled out of the chain index system about a dancer and a pull roller and transferred out of the machine.

Designation	Execution		Comments
Pull roller	Drive roll is driven by a servo motor via gearing and coupling		n.a.
	Nip roll is not actively driven		n.a.
	Nip roll is pulled against the drive roll by cylinders		n.a.
	Nip pressure is adjustable via proportional valve		n.a.
	Opening and closing of station via ASI module with opened guard doors		n.a.
	Ionizing waste grid to remove static		n.a.
	Description/Data		
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
	ionization	See technic. doc.	n.a.
Dancer	Mechanical end stops on both sides of the swivel angle to protect the cylinder and detect torn waste grid		n.a.
	Position control of dancer via sensor		n.a.
	Cylinder for material tension		n.a.
	Pneumatic pressure adjustable via manual pressure regulator		n.a.
	Counter weight for weight compensation for sensitive materials		n.a.
	Description/Data		
	cylinder	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	sensor	See technic. doc.	n.a.
	pressure regulator	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.24 Station 1120, 1121, 1122, 1123 and 1124: Splice detection

The splice in the material is detected by a sensor.

Products around the splice are marked as bad parts in the shift register.

Designation	Execution		Comments
Splice detection	Sensor with holder mounted on aluminum profile		n.a.
	Distance sensor to material is adjustable by the holder according to switching behavior		n.a.
	Sensor adjustable to splice via manual teach-in		n.a.
	Description/Data		
	sensor	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.25 Station 1130 and 1131: Tack sealing

The different materials are sealed together in a small area.

This prevents the materials to shift against each other.

Designation	Execution		Comments
Tack sealing	Tack sealing with cylinders from top and bottom		n.a.
	Sealing tool and counter plate with pluggable heating elements and PT 100		n.a.
	Sealing tool and counter plate with quick release fastener mounted on station		n.a.
	Sealing pressure adjustable via proportional valve		n.a.
	Sealing tool and counter plate isolated from the outside (G60)		n.a.
	Description/Data		
	sealing time	max. 1,0 s	n.a.
	Temp. back up tool	approx.	n.a.
	Temp. sealing tool	approx.	n.a.
	sealing pressure	approx. 2,5 to 3,5 bar	n.a.
	heating elements	See technic. doc.	n.a.
	PT 100	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.26 Station 1140, 1141, 1142, 1143, 1144 and 1145: Web guiding

The web guiding aligns the web to the following stations.

Designation	Execution		Comments
Web guiding	Web guiding Interface with HH control via ProfiNet		n.a.
	Web position regulation via edge guiding (one sensor)		n.a.
	Regulation by rotating frame		n.a.
	Correction of web position during machine run via HMI possible		n.a.
	Description/Data		
	rotating frame	See technic. doc.	n.a.
	web guiding	See technic. doc.	n.a.
	sensor	See technic. doc.	n.a.
	accuracy of web guiding	+/- 0,25mm	n.a.
Programming instructions		n.a.	

7.3.2.27 Station 1146: Web guiding

The web guiding aligns the top web to the web in the chain in the Y-direction. For aligning the webs in the X-direction one idle roller is adjustable via servo motor.

Designation	Execution		Comments
Web guiding	Web guiding Interface with HH control via ProfiNet		n.a.
	Web position regulation via edge guiding (one sensor)		n.a.
	Regulation by rotating frame		n.a.
	Adjusting unit for automatic X-adjustment via servo motor		n.a.
	Correction of web position during machine run via HMI possible		n.a.
	Description/Data		
	rotating frame	See technic. doc.	n.a.
	web guiding	See technic. doc.	n.a.
	sensor	See technic. doc.	n.a.
	accuracy of web guiding	+/- 0,25mm	n.a.
	servo motor	See technic. doc.	
Programming instructions		n.a.	

7.3.2.28 Station 1150, 1151, 1152, 1153, and 1154: Unwinding

By means of a driven expansion reel core holder the material is unwound from the master reel into a dancer. Unwinding is controlled via the dancer position.

The splicing table makes it easier to splice the materials.

Material tears and foil end are detected via dancer position.

Designation	Execution		Comments
Unwinding	Unwinding of material by servo motor via gearing		n.a.
	Pneumatic expansion reel core holder with internal air supply		n.a.
	End of roll is calculated		n.a.
	Opening and closing of reel core holder via hand valve with opened guard doors		n.a.
	Description/Data		
	unwinding	continuous	n.a.
	max. foil width	See chapter 6.1.	n.a.
	max. roll diameter	See chapter 6.1.	n.a.
	diameter reel core holder	See chapter 6.1.	n.a.
	max. roll weight	See chapter 6.1.	n.a.
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
	valve	See technic. doc.	n.a.
Splice table	Splice table with scale – cross index direction – and recessed cutting line		n.a.
	Splice table with clamping elements		n.a.
	Torn material controlled by dancer position		n.a.
	Opening and closing of clamping via ASI module with opened guard doors		n.a.
	Description/Data		
	clamping elements	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
Dancer	Mechanical end stops on both sides of the swivel angle to protect the cylinder		n.a.
	Position control of dancer via sensor		n.a.
	Cylinder for material tension		n.a.
	Pneumatic pressure adjustable via manual pressure regulator		n.a.

	Counter weight for weight compensation for sensitive materials		n.a.
	Description/Data		
	cylinder	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	sensor	See technic. doc.	n.a.
	pressure regulator	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.29 Station 1170, 1171, 1172, 1173, 1174, 1175, 1176, 1177 and 1178: Pull roller

The foil and nonwoven material is feed into the machine by pull rolls.

Designation	Execution		Comments
Pull roller	Pull roll is driven by a servo motor via gearing and coupling		n.a.
	Nip roll is driven by the drive roll via gear wheels		n.a.
	Nip roll is pulled against the drive roll by cylinders		n.a.
	Nip pressure is adjustable via proportional valve		n.a.
	Opening and closing of station via ASI module with opened guard doors		n.a.
	Description/Data		
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.30 Station 1180, 1181, 1182, 1183, 1184, 1185 and 1186: web tension verification

The web tension is measured by web tension verification and controlled by the pull rolls.

The web tension is adjustable via HMI.

Designation	Execution		Comments
Web tension verification	Station is not adjustable to web position		n.a.
	Tension is measured by two integrated sensors (load cells) in the idle roller		n.a.
	Description/Data		
	Load cells	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.31 Station 1200: Gasket magazine

The gasket magazine is an oval transport system with mandrels. The mandrels are loaded manually.

Designation	Execution		Comments
magazine	32 mandrels for loading gaskets 18 mandrels are accessible for loading		n.a.
	Oval transport is driven by servo motor via gearing		n.a.
	Loading possible while machine is running but transport moves only if guards are closed		n.a.
	Sensors for mandrels in singulation position filled		n.a.
	Description/Data		
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.32 Station 1210 and 1211: Gasket singulation

The gaskets mounted on the mandrels are destacked and singulated.

Designation	Execution		Comments
Singulation	Gaskets are lifted up by a fork driven by servo motor and gearing		n.a.
	Fork is pulled down by cylinder for moving back in home position		n.a.
	Top gasket is picked by vacuum gripper or parallel gripper		n.a.
	Gripper unit is moved and rotated by cylinders		n.a.
	Gripper places gasket on conveyor		n.a.
	Description/Data		
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	
	sensors	See technic. doc.	n.a.
	cylinder	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
Programming instructions		Fork for lifting gasket stack moves down a little after each cycle	

7.3.2.33 Station 1214 and 1215: PAD Conveyor

Singulated gaskets and barriers are transported to the next stations by the PAD conveyor.

Designation	Execution		Comments
Conveyor	PAD conveyor for transporting barriers and gaskets		n.a.
	Driven by servo motor via chain		n.a.
	Description/Data		
	Servo motor	See technic. doc.	n.a.
	conveyor	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.34 Station 1218 and 1219: Double stack detection

The station checks if there are two barriers on top of each other on the conveyor.

Designation	Execution		Comments
Double stack detection	Sensor detects a double stack via lever		n.a.
	Only barriers can be checked		n.a.
	For gaskets the station is lifted by a cylinder so the parts can move through		n.a.
	Description/Data		
	sensor	See technic. doc.	n.a.
	cylinder	See technic. doc.	n.a.
	valve	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.35 Station 1222 and 1223: PAD detection

Camera checks the position and orientation of the PAD and passes the information to the robot control.

Designation	Execution		Comments
Camera	Camera checks the position and orientation of the PAD (information is passed on to robot control)		n.a.
	Camera from above the pickup position		n.a.
	Illumination as ring around pick up position		n.a.
	Description/Data		
	vision details	See vision FDS	n.a.
Programming instructions		n.a.	

7.3.2.36 Station 1226 and 1227: PAD transfer

The PADs are picked up from the conveyor belt and placed into the fixture of the PAD sealing station.

Designation	Execution		Comments
Robot	Places PAD with the right orientation onto fixture of PAD sealing (position and orientation is transmitted by vision system and the robot controlled accordingly)		n.a.
	Description/Data		
	robot	See technic. doc.	n.a.
Format parts	Format parts are designed for Rapid Change Over		n.a.
	PADs are picked up by vacuum gripper		n.a.
	Description/Data		
	valves	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.37 Station 1250 and 1251: Barrier infeed conveyor

Operator places barrier stacks on conveyor. Barriers are transported into singulation position.

Designation	Execution		Comments
Conveyor	Conveyor for feeding barriers into machine		n.a.
	Driven by servo motor via chain		n.a.
	Description/Data		
	servo motor	See technic. doc.	n.a.
	conveyor	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.38 Station 1254 and 1255: Barrier detection

Camera detects stack in position and transmits the information to the robot.

Designation	Execution		Comments
Camera	Camera checks if the stack is in the right position		n.a.
	Camera from above the pickup position		n.a.
	Illumination from above		n.a.
	Description/Data		
	Vision details	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.39 Station 1258 and 1259: Barrier singulation

Barriers are destacked by a robot and placed onto the PAD conveyor.

Designation	Execution		Comments
Barrier singulation	Singulation by robot		n.a.
	Description/Data		
	robot	See technic. doc.	n.a.
Format parts	Format parts are designed for Rapid Change Over		n.a.
	Barriers are destacked by vacuum gripper		n.a.
	Description/Data		
	valves	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.40 Station 1300 and 1301: Splice detection filter

The splice in the filter material is detected by a sensor.

Products in the area of the splice are marked as bad parts in the shift register.

Designation	Execution		Comments
Splice detection	Sensor with holder mounted on aluminum profile		n.a.
	Distance sensor to material is adjustable by the holder according to switching behavior		n.a.
	Sensor adjustable to splice via manual teach-in		n.a.
	Description/Data		
	sensor	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.41 Station 1304 and 1305: Vision system reject marking

Reject marks in AF300 filter are detected and saved in the shift register.

Designation	Execution		Comments
Camera	Camera checks for reject marking		n.a.
	Light from the other side		n.a.
	Description/Data		
	Vision details	See vision FDS	n.a.
Programming instructions		n.a.	

7.3.2.42 Station 1312 and 1313: Pull roller filter

Two tractor rollers pulls the AF300 filter material into punching station. One tractor is in front one behind the punching station.

Designation	Execution		Comments
Pull roller	Tractor rollers driven by servo motor via toothed belt		n.a.
	Guidance for filter material can be moved manually to allow better access		n.a.
	Tractors have separate servo motors to adjust the tension between the tractors		n.a.
	Description/Data		
	servo motor	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.43 Station 1316 and 1317: Punch and transfer filter

Filters are punched and transferred into sealing station

Designation	Execution		Comments
Punch	Punching tool male / female die		n.a.
	Punch is driven by cylinder		n.a.
	Description/Data		
	cylinders	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
Transfer	Filter is lifted out of punching tool by vacuum gripper		n.a.
	Gripper is moved up and down by cylinder		n.a.
	Horizontal movement driven by servo motor via gearing		n.a.
	Description/Data		
	cylinders	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
	servo motor	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.44 Station 1350 and 1351: Unwinding filter

By means of a driven expansion reel core holder the material is unwound from the master reel into a dancer. Unwinding is controlled via the dancer position.

The splicing table makes it easier to splice the materials.

Material tears and foil end are detected via dancer position and rpm of unwinding.

Designation	Execution		Comments
Unwinding	Unwinding of material by servo motor via gearing		n.a.
	Pneumatic expansion reel core holder with internal air supply		n.a.
	End of roll is calculated		n.a.
	Opening and closing of reel core holder via hand valve with opened guard doors		n.a.
	Description/Data		
	unwinding	continuous	n.a.
	max. foil width	See chapter 6.1.	n.a.
	max. roll diameter	See chapter 6.1.	n.a.
	diameter reel core holder	See chapter 6.1.	n.a.
	max. roll weight	See chapter 6.1.	n.a.
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
	valve	See technic. doc.	n.a.
Splice table	Splice table with scale – cross index direction – and recessed cutting line		n.a.
	Splice table with clamping elements		n.a.
	Torn material controlled by dancer position		n.a.
	Opening and closing of clamping via ASI module with opened guard doors		n.a.
	Description/Data		
	clamping elements	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
Dancer	Mechanical end stops on both sides of the swivel angle to protect the cylinder		n.a.
	Position control of dancer via sensor		n.a.
	Cylinder for web tension		n.a.
	Pneumatic pressure adjustable via manual pressure regulator		n.a.
	Description/Data		

	cylinder	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	sensor	See technic. doc.	n.a.
	pressure regulator	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.45 Station 1380 and 1381: Rewinding waste grid filter

The waste grid is pulled through a dancer and a pull roller and transferred to the rewind system executed as a friction rewind shaft.

Designation	Execution		Comments
Pull roller	Drive roll is driven by a servo motor via gearing and coupling		n.a.
	Nip roll is not actively driven		n.a.
	Nip roll is pulled against the drive roll by cylinders		n.a.
	Nip pressure is adjustable via proportional valve		n.a.
	Opening and closing of station via ASI module with opened guard doors		n.a.
	Description/Data		
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
Dancer	Mechanical end stops on both sides of the swivel angle to protect the cylinder and detect torn waste grid		n.a.
	Position control of dancer via sensor		n.a.
	Cylinder for web tension		n.a.
	Pneumatic pressure adjustable via manual pressure regulator		n.a.
	Description/Data		
	cylinder	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	sensor	See technic. doc.	n.a.
	pressure regulator	See technic. doc.	n.a.
Rewinding	Pneumatic friction shaft with central air supply (rotary distributor)		n.a.
	Pneumatic adjustment of friction force		n.a.
	Proportional valve for friction force control		n.a.
	Servomotor and gear		n.a.
	Control of rewinding speed and friction shaft via diameter sensor		n.a.
	Reel full detection with pre-warning		n.a.
	Control of rewinding speed and friction shaft via diameter sensor		n.a.

	Reel full detection with pre-warning		n.a.
	Description/Data		
	rewinding	continuous	n.a.
	max. foil width	xxxmm	n.a.
	max. roll diameter	xxxmm	n.a.
	diameter reel core holder	3”	n.a.
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
	valve	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.46 Station 1400, 1410, and 1420: Transfer labels

The labels are picked up from the labeler via robot and placed on the bag.

Designation	Execution		Comments
Robot	Robot picks up labels from a labeler and places them with the right orientation onto the bag (position and orientation are transmitted by vision system and the robot controlled accordingly)		n.a.
	Labels are picked up by vacuum gripper		n.a.
	Description/Data		
	robot	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.47 Station 1402, 1412, and 1422: Detection labels (vision system)

The vision system checks the position and orientation of the labels and passes the information to the robot control.

Designation	Execution		Comments
Camera	Camera checks the position and orientation of the PAD (information is passed on to robot control)		n.a.
	Cameras from above or beneath depending on pickup position		n.a.
	Illumination as ring around pick up position		
	Description/Data		
	Vision details	See vision FDS	n.a.
Programming instructions		n.a.	

7.3.2.48 Station 1405, 1415, and 1425: Feeding labels

The liner is indexed by pull rolls. Via a label dispensing unit, the labels released of the liner and handed over to the vacuum pick head of the robot.

Designation	Execution		Comments
Pull roller	Pull roll is driven by a servo motor via gearing and coupling		n.a.
	Nip roll is driven by the drive roll via gear wheels		n.a.
	Nip roll is pulled against the drive roll by cylinders		n.a.
	Nip pressure is adjustable via proportional valve		n.a.
	Opening and closing of station via ASI module with opened guard doors		n.a.
	Sensor to control the index. Sensor is detecting the label. Sensor position as close as possible to the label dispensing position		n.a.
	Description/Data		
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
Label dispensing unit	Servo driven dispensing unit		n.a.
	Description/Data		
	Servo motor	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
Programming instructions	xxx		

7.3.2.49 Station 1407, 1417 and 1427: Unwinding labels

By means of a driven expansion reel core holder the material is unwound from the master reel into a dancer. Unwinding is controlled via the dancer position.

The splicing table makes it easier to splice the materials.

Loading and splicing of new reels is possible during running machine. Therefor a buffer function is integrated in the dancer.

Material tears and foil end are detected via dancer position.

Designation	Execution		Comments
Unwinding	Unwinding of material by servo motor via gearing		n.a.
	Pneumatic expansion reel core holder with internal air supply		n.a.
	End of roll is calculated		n.a.
	Opening and closing of reel core holder via hand valve with opened guard doors		n.a.
	Description/Data		
	unwinding	continuous or intermediate	n.a.
	max. foil width	See chapter 6.1.	n.a.
	max. roll diameter	See chapter 6.1.	n.a.
	diameter reel core holder	See chapter 6.1.	n.a.
	max. roll weight	See chapter 6.1.	n.a.
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
	valve	See technic. doc.	n.a.
Splice table	Splice table with approx. 10° recessed cutting line		n.a.
	Splice table with clamping elements		n.a.
	Torn material controlled by dancer position		n.a.
	Opening and closing of clamping via ASI module with opened guard doors		n.a.
	Description/Data		
	clamping elements	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
Dancer	Linear dancer, horizontal orientated		n.a.
	Mechanical end stops on both sides to protect the cylinder		n.a.
	Position control of dancer via sensor		n.a.
	Cylinder for material tension		n.a.

	Pneumatic pressure adjustable via pressure regulator Left and righthand side independent		n.a.
	Description/Data		
	cylinder	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	sensor	See technic. doc.	n.a.
	pressure regulator	See technic. doc.	n.a.
	Size of buffer	5 min	n.a.
Programming instructions		n.a.	

7.3.2.50 Station 1408, 1418 and 1428: Rewinding liner

By means of a driven reel core holder, executed as a friction rewind shaft, the liner is rewinded.

Changing of a rewind roll is possible during running machine.

Designation	Execution		Comments
Rewinding	Pneumatic friction shaft with central air supply (rotary distributor)		n.a.
	Pneumatic adjustment of friction force		n.a.
	Proportional valve for friction force control		n.a.
	Servomotor and gear		n.a.
	Control of rewinding speed and friction shaft via diameter sensor		n.a.
	Reel full detection with pre-warning		n.a.
	Control of rewinding speed and friction shaft via diameter sensor		n.a.
	Reel full detection with pre-warning		n.a.
	Description/Data		
	rewinding	continuous	n.a.
	max. foil width	xxxmm	n.a.
	max. roll diameter	xxxmm	n.a.
	diameter reel core holder	3”	n.a.
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
	valve	See technic. doc.	n.a.
Programming instructions		n.a.	

7.3.2.51 Station 1550 and 1551: Unwinding flap

By means of a driven expansion reel core holder the material is unwound from the master reel into a dancer. Unwinding is controlled via the dancer position.

The splicing table makes it easier to splice the materials.

Laoding and splicing of new reels is possible during running machine. Therefor a buffer function is integrated in the dancer.

Material tears and foil end are detected via dancer position.

Designation	Execution		Comments
Unwinding	Unwinding of material by servo motor via gearing		n.a.
	Pneumatic expansion reel core holder with internal air supply		n.a.
	End of roll is calculated		n.a.
	Opening and closing of reel core holder via hand valve with opened guard doors		n.a.
	Description/Data		
	unwinding	continuous	n.a.
	max. foil width	See chapter 6.1.	n.a.
	max. roll diameter	See chapter 6.1.	n.a.
	diameter reel core holder	See chapter 6.1.	n.a.
	max. roll weight	See chapter 6.1.	n.a.
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
	valve	See technic. doc.	n.a.
Splice table	Splice table with scale – cross index direction – and recessed cutting line		n.a.
	Splice table with clamping elements		n.a.
	Torn material controlled by dancer position		n.a.
	Opening and closing of clamping via ASI module with opened guard doors		n.a.
	Description/Data		
	clamping elements	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
Dancer	Linear dancer		n.a.
	Mechanical end stops on both sides to protect the cylinder		n.a.
	Position control of dancer via sensor		n.a.
	Cylinder for material tension		n.a.

	Pneumatic pressure adjustable via pressure regulator Left and righthand side independent		n.a.
	Description/Data		
	cylinder	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	sensor	See technic. doc.	n.a.
	pressure regulator	See technic. doc.	n.a.
	Size of buffer	x min	n.a.
Programming instructions		n.a.	

7.3.2.52 Station 1580 and 1581: Rewinding waste grid flap

The waste grid is pulled through a dancer and a pull roller and transferred out of the machine.

Designation	Execution		Comments
Pull roller	Drive roll is driven by a servo motor via gearing and coupling		n.a.
	Nip roll is not actively driven		n.a.
	Nip roll is pulled against the drive roll by cylinders		n.a.
	Nip pressure is adjustable via proportional valve		n.a.
	Opening and closing of station via ASI module with opened guard doors		n.a.
	Description/Data		
	valves	See technic. doc.	n.a.
	cylinders	See technic. doc.	n.a.
	limit switches	See technic. doc.	n.a.
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
Dancer	Mechanical end stops on both sides of the swivel angle to protect the cylinder and detect torn waste grid		n.a.
	Position control of dancer via sensor		n.a.
	Cylinder for material tension		n.a.
	Pneumatic pressure adjustable via manual pressure regulator		n.a.
	Counter weight for weight compensation for sensitive materials		n.a.
	Description/Data		
	cylinder	See technic. doc.	n.a.
	valves	See technic. doc.	n.a.
	sensor	See technic. doc.	n.a.
	pressure regulator	See technic. doc.	n.a.
Rewinding	Pneumatic friction shaft with central air supply (rotary distributor)		n.a.
	Pneumatic adjustment of friction force		n.a.
	Proportional valve for friction force control		n.a.
	Servomotor and gear		n.a.
	Control of rewinding speed and friction shaft via diameter sensor		n.a.
	Reel full detection with pre-warning		n.a.

	Control of rewinding speed and friction shaft via diameter sensor		n.a.
	Reel full detection with pre-warning		n.a.
	Description/Data		
	rewinding	continuous	n.a.
	max. foil width	xxxmm	n.a.
	max. roll diameter	xxxmm	n.a.
	diameter reel core holder	3"	n.a.
	servo motor	See technic. doc.	n.a.
	gearing	See technic. doc.	n.a.
	sensors	See technic. doc.	n.a.
	valve	See technic. doc.	n.a.
Programming instructions		n.a.	